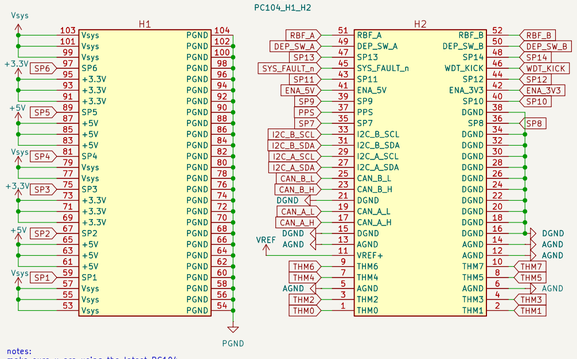
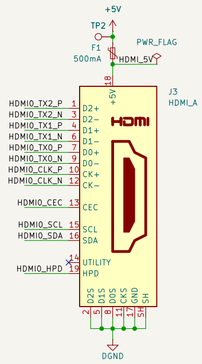
PC104 connectors



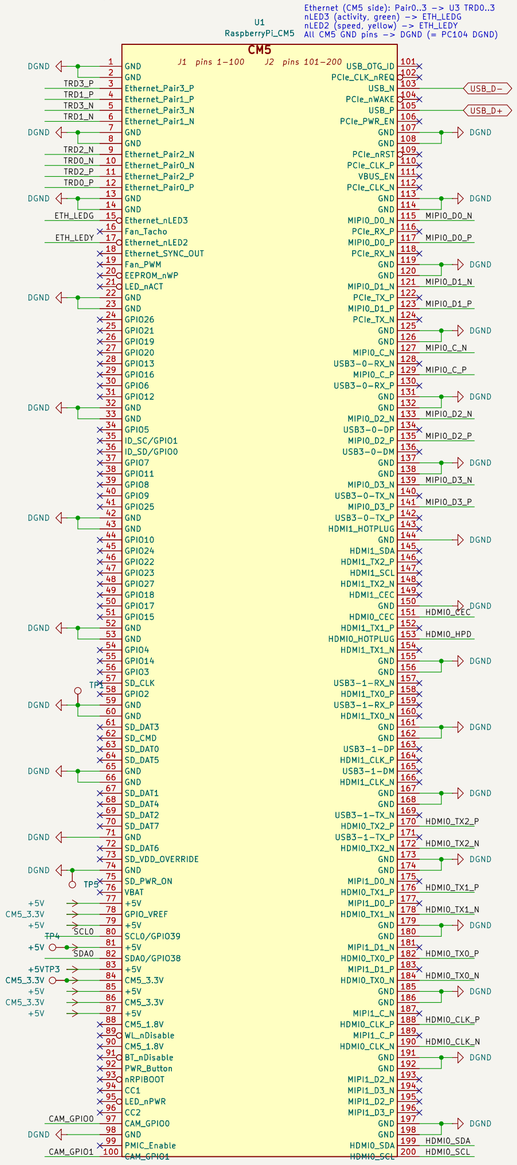
notes:
make sure u are using the latest PC104
Please use the same net label neames as used here
SP = spare

HDMI0



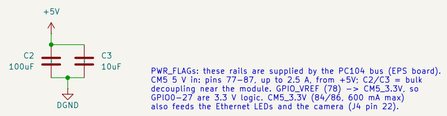
HDMI0 -> HDMI A receptacle J3. TMDS: 100R differential pairs, intra-pair match 0.15 mm. CEC / HPD / DDC connect directly (CM5 has internal pull-up / pull-down, 5 V tolerant). No ESD parts on CM5.
J3 pin 18 (+5V to sink) from +5V rail via F1 500 mA polyfuse.
Pin 14 UTILITY not used. Shields / DDC GND -> DGND.

Raspberry Pi CM5



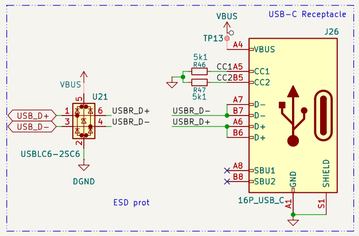
Unused CM5 pins are marked no-connect. USB 2.0 (103/105) goes to the USB-C block on sheet 2.
nRPIBOOT (93) is NC -> for rpiboot eMMC programming pull it low with a jumper, or program on a CM5 IO board.
VBAT (76) NC -> no RTC backup. WL/BT_nDisable NC -> use a non-wireless CM5.

Power

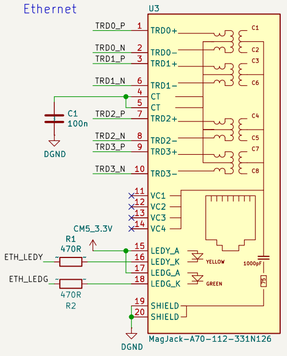


PWR_FLAGs: these rails are supplied by the PC104 bus (EPS board).
CM5 5 V in: pins 77–87, up to 2.5 A, from +5V; C2/C3 = bulk decoupling near the module. GPIO_VREF (78) -> CM5_3.3V, so GPIO0–27 are 3.3 V logic. CM5_3.3V (84/86, 600 mA max) also feeds the Ethernet LEDs and the camera (J4 pin 22).

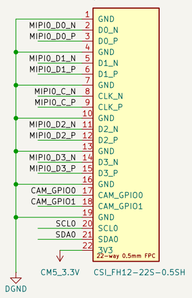
USB-C Receptacle



Ethernet



CSI camera (MIPI0)



J4 = Hirose FH12-22S-0.5SH, 22-way 0.5 mm FPC; pinout = CM5IO CAM/DISP0 (PI 5 camera cable). MIPI0 D0..D3 + clock. SDA0/SCL0 (1.8k pull-ups inside CM5). CAM_GPIO0 = camera power enable. CAM_GPIO1 spare, CM5_3.3V on pin 22. D-PHY pairs 100R diff, 0.15 mm.

Sheet: /
File: main_payload.kicad_sch

Title:

Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 1/1